

Referee Report for “Back to the 1980s or Not? The Drivers of Inflation and
Real Risks in Treasury Bonds”
FINEC-D-24-00175

1 Summary

Campbell, Pflueger, Viceira (CPV, 2020) documented that bond-stock betas changed sign in early 2000s—they were positive in earlier years and became negative in recent years. They developed an asset pricing model with habit formation and macroeconomic ingredients to provide an explanation for this switch, based on a change in the correlation between the output gap and inflation (which was negative in earlier years and became positive in recent years).

While CPV made important contributions, it assumed reduced form dynamic relations for the key variables such as the inflation and the interest rate and left a more structural interpretation for future work. This paper fills this gap by extending CPV with a structural relationship for inflation (the Phillips curve) and the policy interest rate (a Taylor-like policy rule). The paper then applies the model to investigate the drivers of bond-stock betas as well as bond and stock returns. The main finding is that changing bond-stock betas from the 1980s to the 2000s are explained by a change from dominant supply (inflation) shocks to dominant demand shocks, and from a fast and inflation-focused monetary policy rule to a more inertial and output-centric rule.

2 Evaluation

While the paper is interesting, I thought it does not clear the bar for a top finance journal for two main reasons.

1. **Lack of methodological novelty.** The paper is too close to CPV. The model is mostly the same: in fact, most of the equations in Section 2 are the same as in CPV. The difference is the Phillips curve and the monetary policy rule described in Sections 2.4 and 2.5. Methodologically, this is a relatively minor extension. Moreover, the structural additions are either not really structural or not very convincing. Specifically:
 - The Taylor rules are not structural relations. In practice, central banks set policy with discretion to achieve their mandates; they do not follow Taylor rules (although they might use them as benchmark). These rules are best interpreted as reduced form relations that approximate central bank behavior.
 - The Phillips curve is generally the weakest part of New Keynesian models since there is much we don’t understand about inflation (we have a *qualitative* understanding). This paper’s Phillips curve is no exception. The paper assumes prices are flexible and wages are sticky, and derives a wage Phillips curve with backward and forward looking elements. This approach will miss

a lot of richness that matters in practice, e.g., the Covid-19 shock showed that price and wage inflation can have very different dynamics.

To be clear, I like simple models. However, the model should be constructed with a goal of explaining a particular set of facts. We can then evaluate the model based on how well it explains the facts and whether the simplifying assumptions are justified. This gets me to my second point.

2. **The paper does not persuasively explain any new fact (beyond what we can tell based on CPV).** Given the lack of methodological novelty, the paper should be evaluated in terms of how well it explains the motivating facts—specifically, the changing bond-stock betas. The paper claims that betas were positive in the earlier decades because “supply shocks” were more relevant and the Fed was more inflation focused and more alert (less inertial). This explanation is problematic for a number of reasons:

- A qualitative reading of history suggests that there were no major “supply shocks” in the 1980s (either empirically or “in investors’ minds”). Supply shocks were relevant in the 1970s—think oil/food price shocks and associated recessions. In the 1980s, the main issue was elevated inflation and its persistence (e.g., due to inflation expectations or experience affecting price-setting behavior). Chair Volcker hiked rates and generated a recession to reduce inflation. Afterward, there were several “inflation scares”—the markets were testing the Fed’s resolve to see whether they are serious about controlling inflation or whether inflation would flare up again. Over time, the Fed managed to keep inflation at its implicit target (2%). The Fed eventually gained credibility and inflation expectations became anchored. This credibility is why (among other things) bond-stock betas eventually turned negative.¹

Given this history, what should we make of this paper’s explanation that this period featured dominant “supply shocks” with a fast and inflation-focused monetary policy? We can *perhaps* interpret the supply shocks as an “as-if” modeling device to capture inflation scares driven by weak credibility. But we could do a similar interpretation for CPV. This paper’s ambition appears to go deeper than CPV and provide a “structural” interpretation rather than an as-if interpretation. But the structural interpretation does not seem correct.

- A similar qualitative look at the recent decades suggests the Fed is not less focused on inflation and it is not more inertial, compared to the 1980s. As I already mentioned, the Fed managed to keep inflation around 2% for decades until the Covid-19 shock. Inflation flared up recently, but that is because Covid-19 was a large and unusual supply shock. The Fed made a mistake in thinking that this shock was more transitory than it was (the markets made the same mistake throughout 2021). There is no evidence that the Fed was less inflation focused or less inertial. In fact, once the Fed realized its mistake, it quickly reversed course and started a very aggressive hiking

¹This paper has a nice review of this period <https://www.aeaweb.org/articles?id=10.1257/jep.21.4.47>

campaign: raising interest rates by 500 basis points in one year. This seems like *the opposite* of an inertial central bank.

In terms of the behavior of inflation expectations and stock-bond betas in the Covid-19 cycle, there is a simpler explanation than what this paper proposes. The Fed has gained credibility so inflation expectations have not changed very much (beyond a few years). At the same time, because of large macroeconomic shocks (both supply and demand), the Fed had to move interest rates much more than in the past to shift financial conditions and control inflation. The prevalence of macroeconomic shocks increased the interest rate volatility and the bond-stock betas starting from a negative level.² This story does not require a change in the Fed's inertia or inflation focus, and it does not require a complicated model with habit formation.

Overall, while I am sympathetic to the topic and I like the earlier paper by CPV, I cannot recommend this paper to be published in a top finance journal. The paper is relatively minor extension of CPV that does not improve our understanding of the facts much. The paper has quantitative ambitions but the proposed explanation is not persuasive qualitatively. I think the paper should focus on a fact that requires explaining and organize the paper around this fact—constructing a minimalist model with necessary ingredients, estimating the model, and comparing with alternatives.

²The baseline negative bond-stock correlations are likely driven by risk premium or sentiment shocks (or any financial shock that affects stock prices more than bond prices). These shocks are common and they affect stock prices without changing potential output by much. The Fed naturally responds to these shocks in a stabilizing direction, e.g., they cut rates after a stock price decline (the Fed put). This response (and its anticipation) creates a negative correlation between stocks and bonds.